1: Executive Summary

This report presents the design, implementation, and validation of a modular quantitative **risk management and pricing system** used for portfolio analysis, derivative pricing, and market risk measurement. The system integrates three core components: **pricing models, volatility estimation models, and risk measurement frameworks**, all connected through a structured data pipeline and supported by statistical utilities and validation tests.

The primary objective of the system is to provide a consistent framework for evaluating financial risk across equity portfolios and derivative instruments. This includes computing option prices, estimating time-varying volatility, and measuring portfolio risk using industry-standard metrics such as **Value at Risk (VaR)** and **Expected Shortfall (ES)**.

System Overview

The system is organised into four main functional layers:

1. **Pricing Models**
   - Black-Scholes option pricing model
   - Geometric Brownian Motion (GBM) stock calibration
2. **Volatility Models**
   - EWMA volatility estimator
   - GARCH(1,1) volatility model
   - Implied volatility solver (Newton-Raphson method)
3. **Risk Models**
   - Historical VaR and ES
   - Parametric VaR and ES (Variance-Covariance approach)
   - Monte Carlo VaR and ES (stock, option, and portfolio level)
4. **Statistical & Structural Utilities**
   - Covariance matrix estimation
   - Descriptive statistics and distribution diagnostics
   - Backtesting and validation tools

These modules are connected through a structured data pipeline that flows from raw market data ingestion, through preprocessing and volatility estimation, into pricing and finally risk aggregation.

Purpose and Motivation

The system is designed to replicate a simplified but realistic **quantitative risk desk environment**, where models are used for:

- Pricing European options and underlying assets
- Estimating volatility dynamics under changing market conditions
- Measuring downside risk exposure using VaR and ES

- Supporting portfolio-level risk aggregation and diversification analysis

In practical industry applications, these tools are used for:

- Risk reporting and capital allocation
- Trading desk risk monitoring
- Regulatory compliance (e.g., Basel-style risk measures)
- Scenario and stress testing of portfolios

Key Limitations

Despite its modular design, the system has several important limitations:

- **Normality assumptions** in parametric VaR and ES can underestimate tail risk
- Monte Carlo simulations are subject to **sampling error and computational noise**
- Covariance estimation assumes stable linear relationships, which may break down during market stress
- Black-Scholes pricing assumes constant volatility and frictionless markets, which is unrealistic in practice
- Implied volatility estimation depends on numerical convergence and may fail for illiquid or extreme option prices
- The system uses daily data only, limiting high-frequency accuracy and intraday risk dynamics

Additionally, option data and portfolio structures are simplified, and do not fully represent real-world market microstructure effects such as liquidity constraints, transaction costs, or stochastic interest rates.

Overall, the system provides a **conceptually sound and mathematically consistent framework** for pricing and risk analysis in a simplified financial environment. It successfully integrates multiple approaches to volatility estimation, derivative pricing, and risk measurement into a unified architecture.

The inclusion of **VaR, Expected Shortfall, and Monte Carlo simulation frameworks**, combined with **time-varying volatility models (EWMA, GARCH, and implied volatility)**, ensures that the system captures a broad range of market risk dynamics.

However, while the system is **fit for academic and prototype-level risk analysis**, it should be interpreted with caution in real-world applications due to simplifying assumptions around distributional behaviour, volatility dynamics, and market completeness.

2: Introduction

The Risk System is a modular quantitative finance framework designed to support asset pricing, volatility modelling, and portfolio risk measurement. It integrates pricing models, statistical volatility estimators, and risk analytics tools into a unified system for evaluating financial exposure under uncertainty.

<u>Purpose</u>

The purpose of the system is to provide a structured environment for:

- Pricing equity and derivative instruments using standard financial models
- Estimating time-varying volatility using historical and implied approaches
- Measuring portfolio risk using Value at Risk (VaR) and Expected Shortfall (ES)
- Supporting scenario analysis through Monte Carlo simulation

<u>Scope</u>

The system includes:

- Equity and European option pricing (Black-Scholes, GBM calibration)
- Volatility modelling (EWMA, GARCH, implied volatility)
- Risk measurement (historical, parametric, Monte Carlo VaR and ES)
- Statistical and covariance analysis tools

The system excludes:

- Real-time trading execution
- High-frequency intraday modelling
- Full option surface modelling (volatility smiles/skews)
- Complex fixed income or credit derivatives
- Live market risk infrastructure integration

<u>Users</u>

The system is designed for use by:

- Quantitative analysts developing and testing pricing and risk models
- Risk management teams monitoring portfolio exposure and tail risk
- Research and validation teams performing model testing and stress analysis

<u>3: Model Description</u>

The system is composed of three core modelling laters, **pricing models, volatility/statistical models, and then the risk models.** This section describes the theoretical foundation and implementation choices, and limitations of each subsystem

3.1: Pricing Models

The pricing module consists of a **Black-Scholes option pricing model**, **Greeks calculations**, and a **Geometric Brownian Motion (GBM) calibration model** for underlying asset dynamics.

<u>3.1.1: Black-Scholes Option Pricing:</u>

The Black-Scholes model is used to price European call and put options under the following assumptions:

- Asset prices follow a **lognormal distribution**
- Volatility is **constant over time**
- Risk-free rate is **constant**
- Markets are **frictionless** (no transaction costs or arbitrage constraints)
- No early exercise (European-style options only)

The model computes option value using the closed-form solution derived from the risk-neutral valuation framework.

Key limitation: In reality, volatility is stochastic and exhibits clustering and skew, meaning Black-Scholes is an approximation rather than a true market model.

<u>3.1.2 Option Greeks (delta, gamma, vega, theta)</u>

The Greeks module measures sensitivity of option prices to underlying risk factors:

- **Delta** → sensitivity to changes in underlying price
- **Gamma** → curvature of delta (second-order risk)
- **Vega** → sensitivity to volatility changes
- **Theta** → time decay of option value

These are derived analytically from the Black-Scholes formula.

Key role in system:
Greeks are used for:

- Risk exposure monitoring
- Hedging sensitivity analysis
- Understanding nonlinear portfolio behaviour

Limitation:
Greeks assume continuous re-hedging and constant volatility, which is unrealistic in stressed markets.

<u>3.1.3 GBM Stock Calibration</u>

The stock model assumes that asset prices follow a **Geometric Brownian Motion (GBM)** process:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

The calibration function estimates:

- μ (drift) → expected return
- σ (volatility) → standard deviation of returns

These are derived from historical log returns using statistical estimators (MLE or moment-based).

Key assumptions:

- Log returns are normally distributed
- Returns are independent and identically distributed
- Volatility is constant over the estimation window

Key limitation:
Financial returns exhibit:

- Fat tails
- Volatility clustering
- Regime changes
  which violate GBM assumptions, especially during stress periods.

<u>Summary of Pricing Layer</u>

The pricing subsystem provides:

- Closed-form option pricing (Black-Scholes)
- Sensitivity measures (Greeks)
- Statistical calibration of underlying dynamics (GBM)

It is mathematically consistent but relies on **strong simplifying assumptions**, particularly constant volatility and lognormal return distributions. These simplifications are acceptable for a baseline model but are known sources of pricing error in real market conditions.

3.2. Volatility Models

Volatility modelling is a core component of the system because it directly impacts **option pricing, risk measurement, and Monte Carlo simulation accuracy**. Unlike classical pricing assumptions (e.g. Black-Scholes), this system explicitly models volatility as **time-varying**, reflecting real financial market behaviour.

A key assumption of the pricing framework is that **volatility is not constant in practice**, and therefore multiple complementary models are used to capture different aspects of market dynamics.

3.2.1 EWMA Volatility

The EWMA (Exponentially Weighted Moving Average) model estimates volatility using exponentially decaying weights:

- More recent returns receive higher weight
- Older observations decay exponentially
- RiskMetrics standard uses $\lambda = 0.94$

Purpose

- Capture short-term volatility clustering
- Provide responsive but smooth volatility estimates

Key Assumption

- Recent market movements are more relevant than distant history

Strengths

- Simple and computationally efficient
- Adapts faster than rolling window methods
- Widely used in industry risk systems (RiskMetrics framework)

Limitations

- Still backward-looking (historical dependency)
- Can lag during sudden volatility regime shifts
- Does not incorporate market expectations

3.2.2 Garch(1,1) Volatility Model

The GARCH model extends EWMA by explicitly modelling **volatility clustering dynamics**:

$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$

Purpose

- Model persistent volatility clustering
- Capture long-memory effects in financial time series

Key Assumptions

- Returns exhibit conditional heteroskedasticity
- Volatility depends on both:
    - recent shocks (α term)

- past volatility (β term)

Strengths

- More realistic than constant volatility models
- Captures persistence in market volatility
- Standard model in quantitative finance

Limitations

- Parameter sensitivity (ω, α, β must be calibrated carefully)
- Can underperform in regime-switching markets
- Assumes symmetric response to shocks (no leverage effect unless extended)

3.2.3 Implied Volatility

Implied volatility is derived by inverting the Black-Scholes pricing formula using market option prices.

It is solved numerically using the Newton-Raphson method.

Purpose

- Extract market-implied expectations of future volatility
- Used directly in option pricing and risk calibration

Industry Importance

- Implied volatility is the standard in real-world derivatives pricing
- Reflects supply/demand and market sentiment
- Forward-looking (unlike historical models)

Key Assumptions

- Black-Scholes model is used as the pricing baseline
- Market option prices are arbitrage-free and observable

Strengths

- Incorporates market expectations (not just historical data)
- Forward-looking volatility signal
- Essential for pricing derivatives consistently with market data

Limitations

- Dependent on Black-Scholes assumptions (model misspecification risk)

- Numerical instability when Vega is small
- Sensitive to noisy market prices

3.3 Risk Models

The risk modelling layer forms the core of the system's risk management framework. It is responsible for quantifying potential portfolio losses under uncertainty using multiple complementary approaches: Historical VaR, Parametric VaR, Monte Carlo VaR, and Expected Shortfall (CVaR).

A key design principle of this module is that **no single VaR method is sufficient on its own**, because each approach relies on different assumptions about return distributions and market behaviour. Therefore, multiple models are used to provide robustness and cross-validation of risk estimates.

3.3.1 Historical VaR (Non-Parametric Approach)

Historical VaR estimates risk directly from observed past returns without assuming any theoretical distribution.

It is computed by ranking historical losses and taking a percentile of the empirical loss distribution.

Assumptions

- Future risk is similar to historical behaviour
- Returns are representative of current market conditions
- No distributional assumption (fully data-driven)

Key Characteristics

- Fully non-parametric approach
- Captures empirical fat tails and skewness
- Simple and widely used in regulatory reporting

Advantages

- No model assumptions about distribution
- Captures real historical extreme events
- Easy to implement and interpret

Limitations

- Strong reliance on historical data quality
- Fails under structural market regime changes
- Cannot predict unseen extreme events

- Sensitive to sample window selection

3.3.2 Parametric VaR (Variance-Covariance Approach):

Parametric VaR assumes that portfolio returns follow a normal distribution and uses volatility and mean estimates to analytically compute risk.

Assumptions

- Returns are normally distributed
- Portfolio behaves linearly with respect to risk factors
- Volatility is constant over the horizon
- Risk is fully captured by mean and variance

Key Characteristics

- Closed-form solution using Z-scores
- Efficient and computationally fast
- Industry-standard baseline model

Advantages

- Fast and analytically tractable
- Useful for real-time risk reporting
- Easy to extend to portfolios using covariance matrices

Limitations

- Underestimates tail risk due to normality assumption
- Fails for non-linear instruments (e.g. options)
- Poor performance during crisis periods
- Sensitive to volatility estimation errors

3.3.3: Monte Carlo VaR

Monte Carlo VaR simulates thousands of possible future market scenarios using stochastic processes (GBM for stocks and Black-Scholes repricing for options).

Assumptions

- Asset prices follow stochastic processes (GBM dynamics)
- Market evolution can be approximated via random sampling
- Model parameters (μ, σ, r) are known and stable
- Simulated paths represent future uncertainty

Key Characteristics

- Simulation-based risk estimation
- Handles non-linear and path-dependent payoffs
- Extensible to portfolio-level risk aggregation
- Supports stock, option, and combined portfolios

<u>Advantages</u>

- Captures non-linear risk (especially options)
- Flexible and realistic compared to parametric models
- Allows correlation modelling across assets
- Can incorporate complex portfolio structures

<u>Limitations</u>

- Computationally expensive
- Subject to Monte Carlo sampling error
- Highly dependent on model assumptions (GBM, BS)
- Results vary with number of simulations

3.3.4: Expected Shortfall (CVaR)

Expected Shortfall (CVaR) measures the **average loss given that VaR has been exceeded**, providing a more complete view of tail risk.

CVaR = Expected loss conditional on being in the worst α% of outcomes.

<u>Assumptions</u>

- Loss distribution is correctly captured by simulated or historical P&L
- Tail observations are meaningful and stable
- Sufficient data exists in extreme tail region

<u>Key Characteristics</u>

- Coherent risk measure (satisfies subadditivity)
- Focuses explicitly on extreme losses beyond VaR
- Preferred by regulators over VaR in modern frameworks

<u>Why CVaR is used instead of only VaR</u>

- VaR does not describe severity of tail losses
- VaR is not subadditive (can underestimate portfolio risk)
- CVaR provides full tail expectation, not just a threshold
- Better suited for portfolios with options and nonlinear payoffs

<u>Advantages</u>

- Captures tail risk more effectively than VaR
- Mathematically coherent risk measure
- More stable under portfolio aggregation

Limitations

- Requires more data in tail region (less stable in small samples)
- More sensitive to extreme simulation outcomes
- Computationally heavier in Monte Carlo settings

3.4: Covariance Model

The covariance model estimates the statistical relationship between asset returns and is a key component of portfolio risk aggregation. It is implemented through the `covariance_matrix()` function, which computes the covariance matrix of asset returns across multiple assets.

Purpose and Role in the System

The covariance matrix is primarily used for:

- **Portfolio risk aggregation**
- Combining individual asset risks into a single portfolio-level risk measure
- Supporting **parametric VaR and ES calculations**
- Enabling correlation-aware Monte Carlo simulations

In portfolio risk modelling, covariance is essential because it determines how individual asset risks interact rather than assuming independent movements.

Model Definition

The covariance matrix is computed using historical return data:

- Input: time series of asset returns (DataFrame, where columns represent assets)
- Output: an N × N covariance matrix

Each element represents the linear dependence between two assets:

- Positive covariance → assets tend to move together
- Negative covariance → assets move in opposite directions
- Zero covariance → no linear relationship

Assumptions

The covariance model is based on the following assumptions:

- Asset returns are **jointly stationary over the estimation window**
- Relationships between assets are **linear**
- Historical correlations are a valid proxy for future dependencies
- Missing values do not materially distort relationships (handled via `dropna()`)

Key Characteristics

- Computed using standard sample covariance estimation (`numpy.cov`)
- Uses historical return data as the basis for dependency structure
- Forms the backbone of multi-asset risk aggregation
- Directly feeds into portfolio VaR and Monte Carlo simulations

Limitations

Despite its importance, the covariance model has several known weaknesses:

1. Linear Dependence Assumption

The model assumes that relationships between assets are linear, which is often unrealistic in financial markets where:

- Correlations can change non-linearly
- Tail dependencies are not captured

2. Instability in Stress Regimes

During market crises:

- Correlations tend to spike ("correlation breakdown / convergence")
- Historical covariance becomes a poor predictor of future relationships
- Diversification benefits may be overstated

3. Sensitivity to Sample Window

- Short windows → noisy estimates
- Long windows → slow response to regime changes

4. No Regime Switching or Time Variation

- Covariance is assumed static over the estimation period
- Does not capture dynamic correlation structures

The covariance model provides a mathematically consistent and computationally efficient method for estimating asset dependencies and enabling portfolio aggregation. However, its reliance on linear and historically stable relationships makes it less reliable during periods of market stress.

For this reason, it is used in combination with Monte Carlo simulation and multiple VaR methodologies to improve robustness in the overall risk system.

4: Mathematical Framework

This section summarises the key mathematical foundations underlying the pricing, volatility, and risk models implemented in the system.

4.1 Black-Scholes Option Pricing:

The Black-Scholes model is used to price European options under the assumption of lognormally distributed asset prices and constant volatility.

$$C = SN(d_1) - Ke^{-rT}N(d_2)$$

$$D_1 = \frac{ln(S/K) + (r+1/2\sigma^2)T}{\sigma\sqrt{T}}, d_2 = d_1 - \sigma\sqrt{T}$$

This framework assumes:

- Constant volatility
- Constant risk-free rate
- Frictionless markets
- No arbitrage

4.2 Value at Risk (VaR)

VaR measures the maximum expected loss over a time horizon at a given confidence level.

$$VaR_a = \inf\{l: P(L < l) \leq 1 - a\}$$

For parametric VaR under normal returns

$$VaR = V(z_a\sigma\sqrt{T} - \mu T$$

- $Z_a = \Phi^{-1}(a)$
- V is the portfolio value

4.3 Expected Shortfall (CVaR)

Expected Shortfall measures the **average loss beyond VaR**, providing a coherent tail-risk metric.

$$ES_a = E[L \mid L \geq VaR_a]$$

It satisfies **coherence properties**, including subadditivity:

- Preferred over VaR for portfolios with nonlinear payoffs (e.g. options)

4.4: EWMA Volatilty

EWMA models time-varying volatility using exponential decay:

- $\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_{t-1}^2$
- Where
    - $\lambda \in (0,1)$ controls decay (RiskMetrics uses 0.94)
- This implies:
    - Recent returns have higher weight
    - Volatility is not constant over time

4.5 Garch (1,1) Volatility

GARCH models volatility clustering explicitly:

- $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta\sigma_{t-1}^2$
- Where:
    - $\omega > 0$
    - $\alpha, \beta \geq 0$
    - $\alpha + \beta < 1$ for stationarity
- This captures
    - Persistence of volatility shocks
    - Mean-reverting variance dynamics

4.6 Covariance Matrix

Portfolio dependencies are captured using the covariance matrix

- $\Sigma_{ij} = E[(r_i - \mu_i)(r_j - \mu_j)]$

Empirically estimated as:

- $\Sigma = 1/T\ (R - R_{bar})^T)(R - R_{bar})$

This is used for:

- Portfolio aggregation
- Correlated Monte Carlo simulation
- Risk decomposition

5: Implementation and Numerical Methods

This section describes the numerical implementation choices used across the risk system, with emphasis on Monte Carlo simulation design, convergence properties, and computational limitations. The system combines closed-form models (Black-Scholes) with simulation-based methods (Monte Carlo) to evaluate portfolio risk under non-linear payoff structures.

5.1 Monte Carlo Simulation Framework

Monte Carlo simulation is the core numerical method used for portfolio risk estimation, particularly for VaR and Expected Shortfall calculations. It is applied at three levels:

- Single stock VaR (monte_carlo_stock_var)
- Option VaR using re-pricing (monte_carlo_option_var)
- Full portfolio aggregation (monte_carlo_portfolio_var)

Simulation Parameters

The implementation uses the following standard configurations:

- Number of simulations (n_sim):
  - 10,000 for stock-only simulations
  - 50,000 for option and portfolio simulations (higher variance requirement)
- Time horizon:
  - Typically 10 trading days (short-term risk horizon consistent with industry VaR standards)
- Random seed:
  - Fixed seed (seed=42) to ensure reproducibility of results and consistent validation testing

Stock Simulation (GBM-based)

Stock paths are simulated using a Gaussian return model consistent with Geometric Brownian Motion assumptions:

- Returns drawn from a normal distribution
- Scaled by volatility and horizon length
- P&L computed from simulated price changes

Option Simulation

Option risk is estimated by:

1. Simulating underlying stock paths (GBM process)
2. Repricing options at horizon using Black-Scholes
3. Computing P&L as change in option value

This ensures non-linear payoff sensitivity is captured correctly.

5.2 Monte Carlo Numerical Properties

Convergence and Simulation Error

A key property of Monte Carlo methods is that estimation error decreases at the rate:

- MC error $\propto 1 / \sqrt{N}$

where N is the number of simulations.

This implies:

- Increasing simulations improves accuracy
- Diminishing returns occur at higher N
- Option VaR requires higher N due to non-linearity and tail sensitivity

5.3 Numerical Issues and Stability

(1) Implied Volatility Convergence

Implied volatility is solved using the Newton-Raphson method.

Key numerical issues include:

- Convergence failure when Vega is very small
- Instability for deep in-the-money or deep out-of-the-money options
- Sensitivity to initial guess (sigma = 0.3 default)

This reflects a known industry issue where implied volatility is not always uniquely defined for illiquid or extreme strikes.

(2) Simulation Noise

Monte Carlo outputs contain statistical noise due to finite sampling:

- VaR estimates vary across runs (even with fixed parameters)
- Option P&L distributions are highly non-linear
- Tail estimates (99% VaR) are particularly noisy

This is especially relevant for:

- Options (gamma exposure increases variance)
- Portfolio aggregation (correlation amplifies noise effects)

(3) Covariance Sensitivity

The portfolio simulation uses a covariance/correlation structure via Cholesky decomposition.

Key limitation:

- Covariance estimates are highly sensitive to outliers
- Stress periods can distort linear correlation assumptions
- Linear dependence assumption breaks down during market crises

This directly impacts:

- Diversification benefits
- Portfolio VaR accuracy
- Tail risk aggregation

5.4 Performance Considerations

To ensure computational feasibility:

- Vectorised NumPy operations are used throughout simulation
- Pre-allocation of arrays reduces memory overhead
- Cholesky decomposition is computed once per simulation
- Option pricing uses vectorised Black-Scholes evaluation inside loops

Trade-off:

- Higher simulation counts improve accuracy but increase runtime
- Portfolio-level MC (50,000 paths × multiple assets) is computationally intensive

5.5 Summary of Numerical Design Choices

The system prioritises:

- Statistical robustness (Monte Carlo + multiple risk methods)
- Reproducibility (fixed seeds)
- Flexibility (stock, option, and portfolio levels)

However, numerical accuracy is constrained by:

- Finite simulation error
- Model misspecification (Black-Scholes assumptions)
- Instability in implied volatility estimation
- Sensitivity of covariance estimation to extreme observations

Overall, the numerical framework balances realism and computational efficiency. Monte Carlo simulation provides flexibility for non-linear derivatives, while analytical models ensure interpretability and speed. However, all simulation-based outputs must be interpreted with awareness of sampling error, convergence limitations, and structural model assumptions.

6: Validation Methodology and Scope

The validation framework is designed to ensure that all components of the risk system are **mathematically correct, numerically stable, and consistent with financial theory**. Validation is performed across four dimensions: pricing accuracy, volatility stability, risk model correctness, and covariance structure integrity.

The testing approach combines **component-level unit tests, sensitivity analysis, robustness checks, and statistical validation**, reflecting how quantitative models are typically validated in industry risk systems.

6.1: What Was Tested

Pricing Correctness

The pricing layer was validated to ensure that:

- Black-Scholes option prices are finite, positive, and economically consistent
- Option Greeks (delta, gamma, vega, theta) behave smoothly with respect to input changes
- GBM calibration produces valid estimates of drift (μ) and volatility (σ)

Key checks include:

- No NaN or infinite outputs
- Option prices increase with volatility (as expected theoretically)
- Consistency between stock price dynamics and GBM assumptions

Volatility Stability

Volatility models were tested to ensure they produce realistic and stable estimates under different market conditions.

Validated models include:

- EWMA volatility (reactive short-term estimator)
- GARCH(1,1) volatility (persistent volatility clustering model)
- Implied volatility (market-derived forward-looking volatility)

Key validation points:

- Volatility is always positive and finite
- EWMA and GARCH respond correctly to return shocks
- Implied volatility converges under Newton-Raphson iteration (when market prices are well-behaved)

A critical requirement is that **volatility is not constant**, reflecting real financial markets where volatility clusters and evolves over time.

Risk Model Consistency

The risk framework was validated across:

- Historical VaR / ES
- Parametric VaR / ES
- Monte Carlo VaR (stock, option, portfolio)
- Expected Shortfall (CVaR)

Checks included:

- VaR is non-negative and increases with volatility
- CVaR ≥ VaR (tail risk property)
- Monte Carlo outputs are stable under fixed seeds
- Portfolio risk aggregation behaves consistently with diversification effects

This ensures that risk outputs are economically meaningful and internally consistent.

Covariance Structure

The covariance model was tested to ensure:

- Correct computation of asset return covariance matrix
- Symmetry of covariance outputs
- Positive diagonal entries (variance positivity)

It is used for:

- Portfolio aggregation in Monte Carlo simulation
- Correlation modeling between assets

Key limitation: covariance assumes **linear dependence**, which may break down during crisis periods when correlations increase non-linearly.

6.2: Types of Testing

The validation framework is structured into four main categories:

1. Component Testing

Each module is tested independently:

- Pricing models (Black-Scholes, GBM calibration)
- Volatility estimators/ Covariance calculations (EWMA, GARCH, implied vol)

- Risk engines (VaR / ES implementations)

Objective: ensure correct implementation at function level

2. Sensitivity Testing

Used to confirm economic intuition:

- Higher volatility → higher option prices
- Higher volatility → higher VaR / ES
- Time horizon increases → risk increases

This ensures models behave consistently with financial theory.

3. Robustness Testing

Evaluates model stability under stress:

- Extreme returns
- High volatility regimes
- Edge cases ($T \to 0$, deep OTM options)

Key focus areas:

- Monte Carlo convergence stability
- Implied volatility numerical stability
- Tail risk sensitivity

4. Statistical Testing

Ensures outputs are statistically valid:

- Covariance matrix symmetry and positivity
- Distributional checks on returns
- Consistency between analytical and simulation-based models

This ensures that outputs are not only computationally correct but statistically meaningful.

6.3 Link to Test Suite

The validation framework is directly implemented through the project test structure located in:

docs/test_plan/

docs/test_results/

The test suite is divided into modular groups:

- **Group A — Data Layer Tests**
    - Validates data loading, cleaning, and preprocessing
- **Group B — Core System Tests**
    - Validates pipeline execution and configuration consistency
- **Group C — Pricing Model Tests**
    - Validates Black-Scholes pricing, GBM calibration, and Greeks stability
- **Group D — Risk Model Tests**
    - Validates VaR, ES, and Monte Carlo risk aggregation
- **Group F — Volatility & Statistical Tests**
    - Validates EWMA, GARCH, implied volatility, and covariance structure

7. Validation Results

The validation framework was executed across all major system components, including the data layer, pricing models, volatility models, covariance estimation, and portfolio risk models.
All validation tests completed successfully with no critical numerical failures, runtime errors, or structural inconsistencies detected.

The results demonstrate that the system is mathematically stable, internally consistent, and operationally reliable for pricing and risk analysis under the tested scenarios.

7.1 Data Layer Validation Results

Test 1 - Raw Data File Generation

Status: Passed

Validation confirmed that all required raw datasets were successfully generated:

- `stock_prices.csv`
- `portfolio_positions.csv`
- `option_data.csv`
- `rates.csv`

No missing files or corrupted outputs were detected.

Test 2 - Returns Matrix Processing

Status: Passed

The processed returns dataset was successfully generated and validated:

- `returns_matrix.csv` created correctly
- No NaN values remained after preprocessing
- Return matrix contained valid numerical observations

The processed return series was therefore considered mathematically usable for downstream modelling.

Test 3 - Loader Integration

Status: Passed

The loader interface successfully imported:

- prices
- returns
- portfolio data
- option data
- rates

All datasets were non-empty and accessible system-wide.

7.2 Core System Validation Results

Test 1 - Main Pipeline Execution

Status: Passed

The full system executed end-to-end without runtime errors.

Successfully completed:

- Stock VaR simulation
- Option VaR simulation
- Portfolio aggregation
- CVaR calculations

No import failures, crashes, or numerical exceptions occurred.

Test 2 - Configuration Consistency

Status: Passed

Global configuration parameters propagated correctly throughout the system:

- Confidence levels applied consistently
- Monte Carlo horizon applied correctly
- Simulation counts matched configuration settings

Outputs remained internally consistent across all modules.

Test 3 - Portfolio Risk Consistency

Status: Passed

Portfolio aggregation results satisfied logical consistency conditions:

- CVaR ≥ VaR
- Diversification ratio finite and positive
- Portfolio risk metrics remained stable under simulation

No NaN or infinite outputs were detected.

7.3 Pricing Model Validation Results

Test 1 - Black-Scholes Output Validity

Status: Passed

The Black-Scholes pricing model produced:

- Positive option prices
- Finite outputs
- Distinct call and put valuations

No numerical instability was observed.

Test 2 — GBM Calibration Stability

Status: Passed

GBM calibration successfully produced:

- Positive volatility estimates
- Finite drift estimates
- Stable parameter outputs across test datasets

No calibration failures occurred.

Test 3 — Volatility Sensitivity

Status: Passed

Option prices increased monotonically with higher volatility inputs, confirming financially consistent behaviour.

This verified correct implementation of volatility dependence within the pricing framework.

7.4 Risk Model Validation Results

Test 1 — Historical VaR Consistency

Status: Passed

Historical VaR calculations produced:

- Positive VaR values
- CVaR values greater than or equal to VaR
- Correct tail-loss behaviour

Results remained stable across different loss distributions.

Test 2 — Parametric VaR Sensitivity

Status: Passed

The parametric VaR model correctly responded to volatility changes:

- Higher volatility produced larger VaR estimates
- All outputs remained finite and positive
- Scaling behaviour remained mathematically consistent

Test 3 — Monte Carlo Portfolio Coherence

Status: Passed

Monte Carlo portfolio simulations successfully generated:

- Stable P&L distributions
- Finite VaR estimates
- Consistent aggregation behaviour

Portfolio diversification effects were observed as expected.

Simulation outputs remained reproducible under fixed random seeds.

7.5 Volatility & Statistical Model Validation Results

Test 1 — EWMA and GARCH Stability

Status: Passed

EWMA and GARCH volatility models produced:

- Positive volatility estimates
- Stable annualised volatility series
- No NaN or infinite outputs

Both models responded appropriately to changes in return variability.

Test 2 - Implied Volatility Consistency

Status: Passed

The implied volatility solver successfully recovered volatility values from synthetic option prices.

Observed results:

- Numerical convergence achieved
- Implied volatility closely matched true volatility inputs
- No major Newton-Raphson instability detected

Minor instability may still occur for extremely deep out-of-the-money options where Vega becomes very small.

Test 3 - Covariance Matrix Validation

Status: Passed

Covariance matrix outputs satisfied key structural requirements:

- Matrix symmetry verified
- Positive diagonal entries confirmed
- Statistical outputs remained finite

The covariance model therefore behaved consistently under correlated return inputs.

7.6 Overall Validation Summary

Validation Area	Result
Data Pipeline	Passed
Pricing Models	Passed
Volatility Models	Passed

Risk Models	Passed
Monte Carlo Simulation	Passed
Covariance Structure	Passed
Statistical Consistency	Passed

7.7 Observed Numerical Limitations

Although all tests passed successfully, several known modelling and numerical limitations remain:

- Monte Carlo outputs contain simulation noise
- Parametric VaR relies on normality assumptions
- Covariance estimates remain sensitive to outliers
- Implied volatility convergence may weaken for illiquid or deep OTM options
- Historical methods depend heavily on past sample quality

Despite these limitations, the validation results indicate that the system is robust, internally consistent, and suitable for academic quantitative risk analysis and prototype-level portfolio risk modelling.

8. Conclusions & Recommendations

8.1 Summary

The quantitative risk management system successfully integrates pricing models, volatility estimation techniques, and portfolio risk measurement frameworks into a unified and modular architecture. The implemented models were validated through component, robustness, sensitivity, and statistical testing, with all major tests passing successfully.

The system demonstrated:

- Correct implementation of Black-Scholes option pricing
- Stable volatility estimation using EWMA and GARCH models
- Consistent Value at Risk (VaR) and Expected Shortfall (ES/CVaR) calculations
- Valid covariance matrix construction for portfolio aggregation
- Stable Monte Carlo simulation outputs under controlled configurations

The mathematical framework underlying the system is internally consistent and aligns with standard quantitative finance methodologies used in industry risk management environments.

The project therefore provides a conceptually sound and operationally functional framework for:

- Derivative pricing
- Volatility modelling
- Portfolio risk measurement
- Tail-risk analysis
- Portfolio aggregation and diversification analysis

8.2 Limitations

Despite successful implementation and validation, several important limitations remain.

Parametric VaR Assumptions

The parametric VaR model assumes that returns are normally distributed. In practice, financial returns often exhibit:

- Fat tails
- Skewness
- Extreme market shocks

As a result, parametric VaR may underestimate tail risk during stressed market conditions.

Monte Carlo Simulation Noise

Monte Carlo risk estimates are subject to simulation randomness and sampling error. Although increasing the number of simulations improves stability, computational cost also increases significantly.

Monte Carlo error decreases approximately at a rate proportional to:

- MC error $\propto 1 / \sqrt{N}$

where N is the number of simulations.

Covariance Instability

The covariance framework assumes relatively stable linear relationships between asset returns. During market crises, correlations can change rapidly and diversification benefits may collapse.

Covariance estimation is also sensitive to:

- Outliers
- Regime shifts

- Structural market breaks

Black-Scholes Assumptions

The Black-Scholes framework assumes:

- Constant volatility
- Lognormal asset returns
- Frictionless markets
- Continuous hedging

These assumptions are unrealistic in real financial markets where volatility is stochastic and market frictions exist.

Implied Volatility Convergence

The implied volatility solver relies on Newton-Raphson numerical iteration and may fail to converge for:

- Deep out-of-the-money options
- Illiquid market prices
- Extremely low Vega values

8.3 Recommendations

Several improvements could strengthen the realism and robustness of the system.

Use Implied Volatility for Option Pricing

Historical volatility alone is insufficient for realistic derivative pricing because it is backward-looking. Future implementations should continue prioritizing implied volatility, since it reflects forward-looking market expectations and is the industry standard for options pricing.

Expand Stress Testing

Additional stress testing frameworks should be incorporated to evaluate model behaviour under:

- Market crashes
- Correlation breakdowns
- Volatility spikes
- Liquidity stress scenarios

This would improve robustness under extreme market conditions.

Incorporate Fat-Tail Distributions

Future versions of the risk engine should consider replacing normal return assumptions with more realistic heavy-tailed models such as:

- Student-t distributions
- EVT (Extreme Value Theory)
- Jump-diffusion processes

This would improve tail-risk estimation accuracy.

Improve Covariance Robustness

The covariance subsystem could be enhanced using:

- Shrinkage estimators
- Dynamic correlation models
- Robust covariance estimation techniques

These methods would reduce instability during stressed market regimes.

Extend Volatility Modelling

Additional volatility frameworks such as:

- EGARCH
- Stochastic volatility models
- Volatility surface modelling

could improve derivative pricing realism and capture asymmetric market behaviour.

Overall, the system provides a strong foundational implementation of a modern quantitative risk framework. While simplifying assumptions limit real-world applicability, the project demonstrates a clear understanding of financial modelling theory, numerical implementation, portfolio risk analytics, and model validation methodology.

The integration of pricing, volatility estimation, VaR, Expected Shortfall, covariance analysis, and Monte Carlo simulation creates a mathematically coherent and practically relevant risk analysis environment suitable for academic and prototype-level quantitative finance applications

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